

Epic 3: Data Pre-processing

Story 4: Splitting the Dataset into Features and Target Variable

Before training the machine learning models, the dataset is divided into independent variables (X) and the dependent (target) variable (y). This is a crucial preprocessing step because it separates the input features from the output label, enabling the model to learn the relationship between them.

Independent Variables (X)

The independent variables, represented by X, contain the input features used to predict flood occurrence. These features include weather-related parameters such as rainfall, cloud cover, temperature, humidity, and other relevant environmental attributes. Only the selected input features are included in X, as they provide the information required for accurate prediction.

Target Variable (y)

The dependent variable, represented by y, is the output that the model aims to predict. In this project, the Flood (or Class) column is used as the target variable, indicating whether a flood is likely to occur. Each record contains one corresponding target value, making y a single-column output vector.

Creating X and y

The dataset is separated using the following approach:

```
X = dataset.drop('Flood', axis=1)
```

```
y = dataset['Flood']
```

(If your dataset uses the column name Class instead of Flood, replace Flood with Class.)

Importance

- Separates input features from the prediction target.
- Ensures the model learns from relevant weather parameters.
- Prepares the dataset for train-test splitting.
- Improves the efficiency and accuracy of machine learning models.
- Forms the foundation for model training and evaluation.

Outcome

- Successfully separated the dataset into independent variables (X) and the target variable (y).

- Prepared the data for train-test splitting and machine learning model development.
- Ensured the model receives the correct input features and prediction labels for training.

Figure 10: Splitting the dataset into independent variables (X) and the target variable (y) for flood prediction.

```
#split the data into train and test set from our x and y  
#import train_test_split fucntion  
from sklearn.model_selection import train_test_split  
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.25,random_state=10)
```